07 - Prior Preterm Birth and 17-OHPC

Companion reading handout for simultaneous listening.
Course: Medical Prevention of Spontaneous Early Delivery

The story of seventeen-hydroxyprogesterone caproate is a hard lesson in how medicine corrects itself. For years, a woman whose only risk was a previous spontaneous preterm birth — a normal cervix, nothing else amiss — had a drug we believed would protect her next pregnancy. Today she does not. This lecture is how the field arrived there: a treatment that looked like a breakthrough, a second trial that undid it, and the narrow place progesterone still holds for her.

The Woman and Her Risk

Start with who she is, because her risk is real, and it shapes everything that follows. A prior preterm birth is the single strongest predictor of another one, and the size of that risk climbs with her history. A woman whose first two babies came at term carries only about a **five percent** chance of delivering the next one early. One prior preterm birth lifts that to somewhere between **thirteen and twenty-one percent**, depending on where in her sequence it fell. Two prior preterm births push it to around **forty percent**. And if both of those came very early, between twenty-one and thirty-one weeks, her chance of a third early birth reaches about **fifty-seven percent**, better than a coin toss. This is the woman a preventive drug was meant to protect, and that is why the stakes were so high.

The Promise: Meis 2003

The drug arrived with the **Meis trial, in two thousand three**. Women with a prior spontaneous preterm birth were given a weekly injection of **two hundred fifty milligrams** of seventeen-hydroxyprogesterone caproate, into muscle, carried through much of the pregnancy. The result was striking. Recurrent preterm birth before thirty-seven weeks fell from **about fifty-five percent** in the placebo group to **thirty-six percent** with the drug, a relative reduction of roughly a third. The effect looked strong enough that the trial was stopped early, and within a few years that weekly injection had become standard care across the United States.

The Molecule: A Synthetic Ester

Underneath the success sat a quiet problem, and it lived in the molecule itself. Seventeen-hydroxyprogesterone caproate carries the word progesterone in its name, but the molecule is a **synthetic ester**, built in a laboratory, with a different chemical structure and different behavior in the body from the progesterone we make ourselves. Put natural progesterone on uterine muscle and it quiets the muscle and helps hold the cervix closed. Put seventeen-hydroxyprogesterone caproate on that same muscle and it does neither, and in some experiments it makes the muscle more active rather than less. So the biological reason to expect it to hold off labour was never as solid as the familiar name made it feel.

The Undoing: PROLONG

The proof of that gap came from the trial meant to settle the matter. It was called **Prolong, reported in twenty twenty**, and it repeated Meis almost exactly, the same weekly injection, the same entry criteria, a larger and international group of women. It found nothing. Recurrent preterm birth before thirty-five weeks was **eleven point zero percent** with the drug and **eleven point five percent** with placebo. The relative risk came out at **zero point nine five**, which is to say no real difference at all, and the newborn outcomes matched. The trial built to confirm the drug did not confirm it.

Why the Two Trials Disagreed

The interesting question is how one drug, given the very same way, could look like a breakthrough the first time and like nothing the second. The answer is in who was enrolled. Even with identical criteria, the two trials filled up with very different women. In Prolong, the placebo group delivered early at **roughly half the rate** of the original trial. These were, on the whole, lower-risk women, most of them from outside the United States, and only about **two in a hundred** had a short cervix. When the underlying risk is that much lower, a drug has far less room to change anything, and a genuine effect and no effect start to look the same. A treatment can seem powerful in a very-high-risk group and quietly dissolve in a milder one. After Prolong, the regulators moved to pull the drug from the market, and the guidelines stopped recommending it.

Does Vaginal Progesterone Rescue Her?

That leaves an obvious question. If the injection is gone, does vaginal progesterone, the treatment that works so well for a short cervix, rescue this woman, the one with a prior preterm birth and a cervix that is still long? The honest answer is no, and the way we know is worth sitting with. Pool all the trials together and, at first, vaginal progesterone does seem to help her, cutting early birth by something like a third. But separate those trials by quality, and the entire benefit belongs to the small ones, the studies most exposed to bias. In the large, carefully run trials, the effect is simply gone, a **relative risk sitting near one**. When a benefit lives only in the small, weaker studies and disappears in the strong ones, that is the shape of a mirage, not a real treatment effect.

The Rule That Survives

So we come back to the rule that governs this whole subject. **Vaginal progesterone earns its place for the woman with a prior preterm birth only when a short cervix appears.** In that moment she stops being defined by her history and becomes the short-cervix patient the strong evidence actually covers, and the benefit, which concentrates in precisely that group, comes back. That is what the guidelines now say without hedging: the weekly injection is no longer recommended; for a prior preterm birth, follow the cervix over time, and offer vaginal progesterone the moment it shortens.

The Bedside Rule

The story of seventeen-hydroxyprogesterone caproate is the field correcting itself in slow motion — a single positive trial, a molecule that was never truly progesterone, a confirmatory trial that told the truth, and a small-study mirage standing in for the alternative. What survives is a clean and slightly humbling rule to carry to the bedside. **For a woman whose only risk is a prior spontaneous preterm birth, with a cervix that is still long, there is no progesterone that reliably prevents the next early birth.** What you owe her is honest watching of that cervix, and the moment it shortens, the treatment we trust is ready.